

Internship in Physical Organic Chemistry

Week 2 — Part 2: Instrument Training and Sample Preparation

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Goals of this part:

- Understand the architecture of the instrument you will use
- Know what to observe and record during the engineer-led training session
- Prepare your sample correctly, with justified choices of solvent and concentration
- Complete scratch section 2W-a before preparing the sample

1 The instrument

The instrument you will use is a Thermo Scientific Orbitrap Elite hybrid mass spectrometer. Understanding its architecture at a basic level will help you follow the training session, interpret what you see on screen, and describe the instrument correctly in your Experimental chapter.

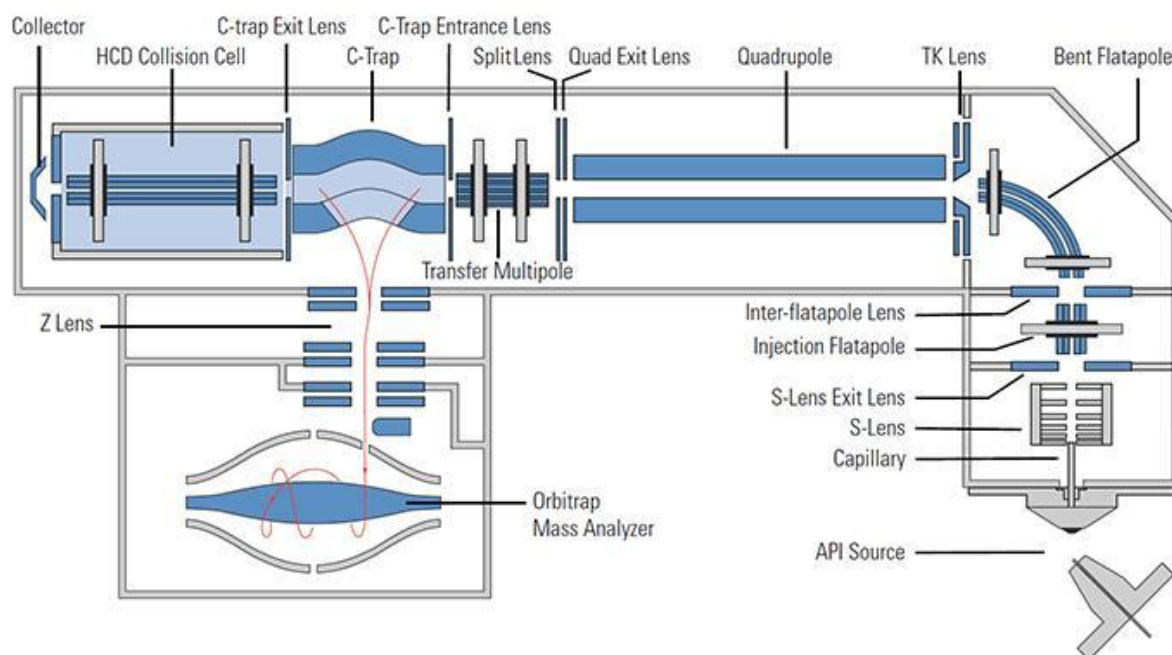


Figure 1: Schematic of the Thermo Scientific Orbitrap Elite mass spectrometer. The ion path proceeds from the API source (bottom right) through the ion transfer optics (capillary, S-lens, flatapoles, quadrupole), into the C-trap, through the HCD collision cell (top left), and into the Orbitrap mass analyser (bottom left). Adapted from Thermo Fisher Scientific.

1.1 Architecture

The Orbitrap Elite is a hybrid mass spectrometer. The diagram provided by your supervisor shows the complete ion path; the description below follows that path from right to left.

The API source (atmospheric pressure ionisation, in your case electrospray, ESI) converts your solution-phase sample into gas-phase ions. A high voltage applied to the spray needle produces a fine mist of charged droplets. As the solvent evaporates, the analyte emerges as singly or multiply charged ions. The polarity of the applied voltage determines whether positive or negative ions are transmitted — this is the step you control when you select positive or negative ion mode.

The ion transfer optics (capillary, S-lens, flatapoles, quadrupole) transmit and focus the ions from the source into the instrument. The quadrupole acts as a mass filter: it can be set to transmit only ions within a narrow m/z window (the isolation window), rejecting all others. This is how the precursor ion is selected for MS/MS experiments.

The C-trap is a curved ion trap that accumulates ions and injects them either into the HCD collision cell for fragmentation, or directly into the Orbitrap for detection.

The HCD collision cell (higher-energy collisional dissociation) is where fragmentation occurs in your experiment. Ions are accelerated through a gas-filled cell and undergo energetic collisions with nitrogen gas molecules. The kinetic energy of these collisions is converted into internal energy, which breaks chemical bonds and produces fragment ions. The amount of energy deposited is controlled by the **normalised collision energy (NCE)** parameter you set in the acquisition method. After fragmentation, the product ions return to the C-trap and are then injected into the Orbitrap.

The Orbitrap mass analyser measures the m/z values of the ions with high resolution and mass accuracy. Ions oscillate around a central spindle-shaped electrode; their oscillation frequencies depend on m/z and are recorded as an image current. A Fourier transform converts this signal into a mass spectrum. The resolving power exceeds 240,000 FWHM, and mass accuracy is within a few ppm.

The key point for your experiment: **precursor selection occurs in the quadrupole, fragmentation occurs in the HCD cell, and detection occurs in the Orbitrap.** These are three distinct, sequential steps on the same instrument.

Connection to the physical organic chemistry concept:

The HCD step is a unimolecular fragmentation reaction with an activation barrier. The NCE parameter controls how much internal energy is deposited into the precursor ion. At low NCE, only low-barrier fragmentation pathways are accessible. At higher NCE, higher-barrier pathways open up — including competing pathways that produce different fragment ions. This is the physical principle behind the energy-resolved measurements you will perform in Week 3: by recording spectra at systematically increasing NCE values, you will map how the relative intensities of fragment ions change as a function of internal energy, and from this you will extract the activation barrier for the retro-Diels-Alder pathway. What you are doing in Week 2 is not just collecting spectra: you are preparing the experimental foundation for a quantitative measurement of activation energy.

1.2 Software

The instrument is controlled by **Xcalibur** (Thermo Scientific). Xcalibur is used to define acquisition methods, monitor the spray in real time, acquire data, and inspect raw spectra. It exports data to Excel. You will use Xcalibur throughout the acquisition days under the supervision of the engineer.

2 The training session

Day 1 of Week 2 is a guided demonstration led by the laboratory engineer. You will observe the engineer operating the instrument. You will not operate it yourself on this day. Your task is to observe carefully, ask questions, and record everything in your lab journal.

2.1 What to observe

During the training session, pay attention to the following, in the order they occur:

1. **Instrument startup and equilibration.** Note how long the instrument takes to stabilise and what the engineer checks before beginning an acquisition.
2. **The ESI source.** Observe the spray needle position, the spray voltage applied, the desolvation gas flow, and the source temperature. Ask the engineer what happens if the spray voltage is too high or too low.
3. **Loading a sample.** Observe how a sample is introduced (syringe pump or direct infusion). Note the flow rate and how the engineer confirms that the sample is reaching the source.
4. **A full-scan MS1 acquisition.** Watch what appears on screen when the instrument scans across the full mass range. Identify the base peak. Ask the engineer how to read the real-time spectrum display.
5. **Precursor selection and MS/MS acquisition.** Observe how a specific m/z is selected as the precursor ion using the quadrupole. Watch the isolation window being set and the NCE value being entered for HCD. Note what changes in the spectrum when the NCE is varied.
6. **Switching between positive and negative ion mode.** Note what the engineer does to switch polarity and how long the transition takes.
7. **Data saving and export.** Observe how raw files are named and saved in Xcalibur, and how data is exported to Excel. Confirm that the file naming convention from Part 1 is compatible with how Xcalibur saves files, and ask the engineer if any adjustments are needed.

2.2 What to record

Use the lab journal template (from Part 1) to record the training session. The objective section should state: *observe instrument operation and learn the acquisition workflow for MS1 and MS/MS experiments in positive and negative ion mode.*

In the observations section, record every instrument parameter you see set by the engineer: spray voltage, source temperature, sheath gas flow, auxiliary gas flow, capillary temperature, resolution setting, isolation window, NCE value(s), scan range, and number of microscans. These parameters will appear in your Experimental chapter. If you do not record them now, you will not be able to report them later.

Be active during your training session. Ask questions. If you do not understand why a parameter is set to a particular value, ask. If you do not know what a displayed number means, ask. The engineer's explanations are part of your data — record them in the lab journal.

3 Sample preparation

You will prepare your own sample before the first acquisition day. Two decisions must be made and justified before you touch any glassware: the solvent and the concentration. Both must be confirmed with your supervisor before you prepare the sample.

3.1 Choosing the solvent

Your choice of solvent must be justified by the literature you read in Week 1. Look at the papers in your Zotero library: what solvent did the authors use to prepare their samples for ESI-MS/MS? Note that the choice is constrained by the instrument: the solvent must be compatible with electrospray ionisation, which requires a volatile organic solvent or a mixture with water. On this instrument, methanol and acetonitrile are the standard choices.

Write your solvent choice and its justification in scratch section 2W-a *before* preparing the sample. Then confirm the choice with your supervisor.

If the papers in your library used a solvent that is not compatible with the instrument, or if different papers used different solvents, record this discrepancy in 2W-a and discuss it with your supervisor. Resolving a discrepancy between literature practice and your instrumental constraints is a normal part of experimental design — it is not a problem to hide.

3.2 Choosing the concentration

A useful concentration range for direct infusion ESI-MS on this instrument is 10^{-5} – 10^{-6} mol L⁻¹. Too concentrated, and the signal will saturate and ionisation will be suppressed. Too dilute, and the signal-to-noise ratio will be insufficient.

Prepare a stock solution at 10^{-4} mol L⁻¹ and dilute it to your working concentration. Keeping a stock solution allows you to prepare fresh working solutions quickly if needed.

Record your target concentration and the preparation procedure in scratch section 2W-a.

3.3 Practical sample preparation

1. Calculate the mass of compound required for your stock solution. Record the calculation in your lab journal.
2. Weigh the compound on an analytical balance. Record the exact mass weighed (not the target mass) in your lab journal.
3. Dissolve in the chosen solvent. Vortex or sonicate briefly if necessary.
4. Prepare the working solution by dilution. Record the dilution in your lab journal with exact volumes used.
5. Transfer to a labelled vial. The label must include: compound name, concentration, solvent, date of preparation, and your name.

Light sensitivity — handling precautions for this project:

Both luteolin and quercetin degrade on exposure to light and should be handled accordingly. Work at a bench away from direct sunlight. Transfer solutions promptly into amber vials, or wrap clear vials in aluminium foil immediately after preparation. When loading the sample into the syringe for infusion, wrap the syringe in aluminium foil as well — the syringe sits in the pump in the open and the solution is exposed throughout the acquisition. Store all solutions in the dark at 4 °C when not in use. Prepare a fresh working solution from the stock on each acquisition day — do not reuse a working solution prepared the previous day. Record the preparation time in your lab journal for each session.

3.4 Verifying the solution

Before proceeding to acquisition, visually inspect your solution. It should be clear and homogeneous. Luteolin and quercetin are yellow in solution; a very pale yellow is normal at the

concentrations used here. If the solution is turbid or shows precipitate, it must not be injected into the instrument. Consult your supervisor.

4 Before the first acquisition day

The following must be completed before you sit down at the instrument for your first acquisition session:

- Scratch section 2W-a filled and confirmed with supervisor
- Sample prepared, labelled, and stored correctly
- Lab journal template printed (one copy per session)
- Folder structure from Part 1 created on your computer
- File naming convention understood and ready to apply

You share the instrument with your partner, who is measuring a different compound. Coordinate your acquisition schedule so that each of you has sufficient instrument time. Your data sets are independent — you each write your own Experimental chapter — but you can help each other during the sessions.